

Caching and accelerating meson builds

Rohit Goswami · 2025-08-06 · cmdline · build

`mold` is an easy win, with the following native file:

```
[binaries]
c_ld = 'mold'
cpp_ld = 'mold'
fortran_ld = 'mold'
[built-in options]
# mold 1.11.0 or later needed due to https://github.com/rui314/mold/issues/1017
# --no-as-needed temporarily necessary due to the seeming return of https://github.com/rui314/mold/issues/1017
c_link_args = ['-fuse-ld=mold', '-Wl,--no-as-needed']
cpp_link_args = ['-fuse-ld=mold', '-Wl,--no-as-needed']
fortran_link_args = ['-fuse-ld=mold', '-Wl,--no-as-needed']
```

`ccache` can be used to cache and rebuild quicker as well:

```
CC="$ (which ccache) $ (which gcc)" CXX="$ (which ccache) $ (which g++)" FCC="$ (which ccache) $ (which gfortran)"
meson setup b2dir --native-file nativeFiles/mold.ini
```

Or in another native file:

```
[binaries]
c = ['ccache', 'gcc']
cpp = ['ccache', 'g++']
fortran = ['ccache', 'gfortran']
```

This can be called in a cascade:

```
meson setup b2dir --native-file nativeFiles/mold.ini --native-file nativeFiles/ccache_gnu.ini
```

Further improvements come from `ccache` with `distcc`, or `icecream` or for a single binary extension, `sccache`.